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OPTIMAL ESTIMATION OF LINEAR OPERATORS IN HILBERT SPACES FROM INACCURATE DATA*

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Abstract. Among all possible methods of estimating a linear operator U from *inaccurate* data, smoothing of the data followed by U is an optimal procedure. This result which we formulate and prove here in a Hilbert space setting may be viewed as an extension of the Golomb–Weinberger method of estimation. It provides as well a rationale for an optimal choice of a smoothing parameter. Our results also provide a method of finding the exact value of the n-width of subsets of a Hilbert space determined by two ellipsoidal bounds. We give an application of this method to the space of essentially time- and band-limited signals introduced by D. Slepian.

1. Introduction. The setting of this paper is adopted from Micchelli and Rivlin [7], which also contains a survey of most of the results in this area. Let U be a linear operator from X into Z, two normed linear spaces. We want to estimate Ux from the following incomplete information about x. A priori information on x is available in the statement that $x \in K$, a convex centrally symmetric subset of X. Also, at our disposal is the approximate value y of a linear information operator I, from X into the normed linear space Y, that is $\|Ix - y\| \leqq \varepsilon$. An estimation algorithm A is then a, not necessarily linear, transformation from $IK + \varepsilon S$, $S = \{y \in Y \mid \|y\| \leqq 1\}$ into Z.

$$\begin{tikzcd} X \arrow[r, "U"] \arrow[d, "I"'] & Z \\ Y \arrow[ur, "A"'] & \end{tikzcd}$$

Since any $x \in K$ and y may be encountered, the error of estimating Ux with Ay is

$$E_A(K, \varepsilon) = \sup_{\substack{x \in K \\ \|Ix - y\| \leqq \varepsilon}} \|Ux - Ay\|$$

and the intrinsic error of the estimation problem is

$$E(K, \varepsilon) = \inf_A E_A(K, \varepsilon). \tag{1.1}$$

Any algorithm achieving this infimum yields an optimal estimation procedure for the linear operator U.

It has been shown [7] that, quite generally, estimating Ux by the Chebyshev center of $H(y) = \{Ux \mid x \in K, \|Ix - y\| \leqq \varepsilon\}$ is an optimal algorithm. The corresponding error satisfies

$$e(K, \varepsilon) \leqq E(K, \varepsilon) \leqq 2e(K, \varepsilon), \qquad e(K, \varepsilon) = \sup_{\substack{x \in K \\ \|Ix\| \leqq \varepsilon}} \|Ux\|. \tag{1.2}$$

However this algorithm is usually neither explicit nor linear. Most of the effort has therefore been directed towards finding explicit optimal *linear* algorithms. It is to be expected that this is not always possible, as is demonstrated by the following simple example.

Example 1.1. Let $X = R^2$, $\|x\|_X^2 = x_1^2 + x_2^2$ with subset $K = \{x \in X \mid \|x\|_X \leqq 1\}$. Let $Ux = x$, taking X into $Z = R^2$, normed by $\|x\|_Z^4 = \lambda_1 x_1^4 + \lambda_2 x_2^4$, $\lambda_1 > \lambda_2 > 0$. Let the

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information operator be $Ix = x_1$, known to within error $0 < \varepsilon < \mu/3$ with $\mu = \sqrt{2\lambda_2/(\lambda_1+\lambda_2)}$. Then the lower bound for the intrinsic error is

$$e^4(K, \varepsilon) = \max(\lambda_2, \lambda_1\varepsilon^4 + \lambda_2(1-\varepsilon^2)^2) = \lambda_2;$$

the latter equality follows from the fact that $\lambda_1\varepsilon^4 + \lambda_2(1-\varepsilon^2)^2 \leqq \lambda_2$ for $\varepsilon \leqq \mu$. However, no linear algorithm will attain this lower bound. Indeed, since any linear algorithm will be of the form $Ay = (\gamma_1 y, \gamma_2 y)$,

$$\begin{aligned} E_A^4(K, \varepsilon) &= \max_{\substack{\|x\|_X \leqq 1 \\ |x_1 - y| \leqq \varepsilon}} \|x - Ay\|_Z^4 \\ &\geqq \max(\lambda_1(1-\gamma_1)^4 + \lambda_2(\gamma_2\varepsilon)^4, \lambda_1(\gamma_1\varepsilon)^4 + \lambda_2(1 \mp \gamma_2\varepsilon)^4) \\ &> \lambda_2, \end{aligned}$$

the first term resulting from $x_1 = y = 1$, $x_2 = 0$, and the second from $x_1 = 0$, $x_2 = 1$, $y = \pm\varepsilon$.

Nevertheless, there is a nonlinear algorithm achieving the lower bound: $Ay = 0$ for $|y| \leqq 2\varepsilon$, $Ay = (y, 0)$ otherwise. The proof rests on the observation that in the first case $|x_1 - y| \leqq \varepsilon$ implies $|x_1| \leqq 3\varepsilon < \mu$, while in the second it implies $|x_1| \geqq \varepsilon$ and hence $x_2^2 \leqq 1 - \varepsilon^2$.

In spite of this example, there are frequent circumstances in which a linear algorithm can be shown to be optimal. One such case occurs when U is a linear *functional*. In this context it has been shown by Micchelli [6], see also Micchelli and Rivlin [7] and in a special case Marchuk and Osipenko [4] that, under rather weak hypotheses, there exists a *linear* functional of the information which is optimal. For the estimation of linear operators however we do not know of any comprehensive result which applies to inaccurate information though in Micchelli and Rivlin [7] several specific instances are analyzed.

On the other hand, when the information is *exact* there is a broadly applicable result in the case that $K = \{x \in X \mid \|x\| \leqq 1\}$ where $\|\cdot\|$ is a Hilbert space semi-norm on X. Then the optimal estimation problem may be viewed as a variant of Golomb and Weinberger's [2] approach, with the result that the optimal estimator consists of applying U to the element in X which minimizes $\|x\|$ subject to $Ix = y$.

The purpose of this paper is to extend this latter result to encompass inaccurate information. Specifically, it will be assumed that Z as well as Y and X are endowed with inner product semi-norms (Example 1.1 points to the necessity of this restriction). We prove that even in the event of error in the information there exists a *linear* algorithm which is optimal and achieves the lower bound (1.2). Our algorithm consists of applying U to the element in X that achieves

$$\min_{x \in X} \|Ix - y\|_Y^2 + \rho(U, \varepsilon)\|x\|_X^2$$

with $\rho(U, \varepsilon)$ a specific nonnegative number depending on U and ε. This prescription, in the case of estimation based upon inaccurate data values $Uf = (f(x_1), \cdots, f(x_m))$ and a priori bound on the nth derivative,

$$K = \left\{ f \,\middle|\, \int_{x_1}^{x_m} \left| f^{(n)}(x) \right|^2 dx \leqq 1 \right\},$$

involves therefore a smoothing of the data, e.g. Reinsch [9], [10] with smoothing parameter depending on the quantity to be evaluated and the error in the data. In

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effect it is the same as the regularization technique employed in the context of ill-posed problems, a point which is also noted by Morozov and Grebennikov [8].

The main theorem used to prove this result, Theorem 2.1, appears to be of interest also in itself. It states that the maximum value of a quadratic form subject to two quadratic inequalities may be found after combining the inequalities linearly in an appropriate fashion. The same theorem may also be used to calculate the Hilbert space n-widths of a set defined by two quadratic inequalities as is done in § 4. This allows us to give an alternate approach to the results of Melkman [5] on n-widths of the space of essentially time- and band-limited signals which was introduced by Slepian in [11]. Unfortunately our results do not generally hold for more than two inequalities. Nevertheless, in § 5 we examine a case in which such an extension is possible. The corresponding optimal estimation problem concerns recovering a function from inaccurate knowledge of some of its Fourier coefficients.

2. A duality formula for nonconvex sets. We begin with the following elementary duality formula.

Let A be a subset of R^2 and define for $a=(a_1, a_2)\in R^2$, $\|a\|_\infty=\max(|a_1|, |a_2|)$, $\|a\|_1=|a_1|+|a_2|$. The inner product of two vectors will be denoted by $\lambda\cdot a$. Also we set $\Delta=\{\lambda\in R^2 \mid \|\lambda\|_1=1\}$.

LEMMA 2.1. *Let A be a convex subset of R^2 with $0\notin$ interior A. Then*

$$\inf_{a\in A}\|a\|_\infty=\max_{\lambda\in\Delta}\inf_{a\in A}\lambda\cdot a.$$

Proof. Let $\delta=\inf_{a\in A}\|a\|_\infty$; then obviously

$$\max_{\lambda\in\Delta}\inf_{a\in A}\lambda\cdot a\leqq\delta.$$

Now the lemma is proved by noting that there is a $\lambda_0\in\Delta$ with $\delta\leqq\lambda_0\cdot a$, $a\in A$. If $\delta=0$ then 0 is a boundary point of A and hence λ_0 clearly exists. If $\delta>0$ then for $0<\varepsilon<\delta$ the convex set $\{a \mid \|a\|_\infty\leqq\delta-\varepsilon\}$ is disjoint from A. Hence there is a $\lambda_\varepsilon\in\Delta$, $b_\varepsilon\in R^1$ with $\lambda_\varepsilon\cdot a\leqq b_\varepsilon$, if $\|a\|_\infty\leqq\delta-\varepsilon$ and $b_\varepsilon\leqq\lambda_\varepsilon\cdot a$ for $a\in A$. Letting $\varepsilon\to 0^+$ through a subsequence, if necessary, we obtain the desired λ_0.

This lemma is a special case of a general duality relation. It holds in infinite dimensional spaces and for norm pairings other than l^∞, l^1 used above.

A proof of this lemma was included here primarily to point out the fact that the essential point in the proof requires separating A from a convex set and not necessarily the convexity of A itself. For our purpose, we wish to maintain the conclusion of Lemma 2.1 but relax the convexity hypothesis on A. Let us set $C(A)=$ convex hull of A and say any subset A of R^2 has a *convex profile* if whenever a supporting hyperplane to A touches A at two points then the line segment joining these points lies in A. In other words, $H\cap A$ is a convex set for any hyperplane H which supports A.

LEMMA 2.2. *A set A has a convex profile if and only if $\partial C(A)\subseteq A$* ($\partial M=$ boundary of M).

Proof. Let $a_0\in\partial C(A)$; then there is a $\lambda\in\Delta$ such that $\lambda\cdot a_0\leqq\lambda\cdot a$ for all $a\in C(A)$. Carathéodory's theorem says that $a_0=\mu_1a_1+\mu_2a_2$, $\mu_1\geqq 0$, $\mu_2\geqq 0$, $\mu_1+\mu_2=1$. If either μ_1 or μ_2 is zero then we are finished. Otherwise it follows that the hyperplane $\lambda\cdot a=\lambda\cdot a_0$ touches at a_1 and a_2 and consequently $a_0\in A$.

Conversely, if $\partial C(A)\subseteq A$ and $\lambda\cdot a=b$ supports A and touches it at two points then any point on the line segment joining the points of contact is also a boundary point of $C(A)$. Hence this line segment must be in A.

Obviously, a set may have a convex profile but not be convex. For instance, A could be an annulus or two positive rays with a common vertex at the origin.

Nevertheless, we have

PROPOSITION 2.1. *Let A have a convex profile. Then the duality relation of Lemma 2.1 remains valid, i.e.,* $\inf_{a\in A}\|a\|_\infty = \max_{\lambda\in\Delta}\inf_{a\in A}\lambda\cdot a$.

Proof. According to Lemma 2.1, we have

$$\inf_{a\in C(A)}\|a\| = \max_{\lambda\in\Delta}\inf_{a\in C(A)}\lambda\cdot a. \tag{2.1}$$

Since, obviously $\inf_{a\in C(A)}\lambda\cdot a = \inf_{a\in A}\lambda\cdot a$ for all $\lambda\in\Delta$, (2.1) simplifies to

$$\inf_{a\in C(A)}\|a\| = \max_{\lambda\in\Delta}\inf_{a\in A}\lambda\cdot a. \tag{2.2}$$

Also,

$$\inf_{a\in A}\|a\|_\infty \geqq \inf_{a\in C(A)}\|a\|_\infty = \inf_{a\in\partial C(A)}\|a\|_\infty \geqq \inf_{a\in A}\|a\|_\infty$$

and thus

$$\inf_{a\in A}\|a\|_\infty = \inf_{a\in C(A)}\|a\|_\infty.$$

Combining this equation with (2.2) gives us the desired conclusion.

The main application we have in mind for Proposition 2.1 is the following situation. Let X be a linear space and suppose $\|x\|_i^2$, $i=0,1,2$ are three Hilbert space semi-norms defined on X. We define

$$B = \{(\|x\|_1^2, \|x\|_2^2) \mid \|x\|_0^2 = 1\}. \tag{2.3}$$

LEMMA 2.3. *B as defined above is an arcwise connected subset of R^2 which has a convex profile.*

Proof. Let $a(x) = (\|x\|_1^2, \|x\|_2^2)$, and suppose $a(x), a(y)\in B$. Define for *all* $t\in R$ $a(t) = a(Zt))$, $Z(t) = (tx+(1-t)y)/\|tx+(1-t)y\|_0$. Note that since $\|x\|_0^2 = \|y\|_0^2 = 1$ then $|(x,y)_0|\leqq 1$ and hence $\|tx+(1-t)y\|_0 = 0$ if and only if $t=\frac{1}{2}$ and $(x,y)_0 = -1$. Let us first assume $(x,y)_0 \neq -1$. Then $a(t)$ is a continuous curve lying in B joining $a(x)$ to $a(y)$. Moreover, if $\lambda\cdot a \geqq b$ is a half-space supporting B and touching it at $a(x)$ and $a(y)$ then the quadratic polynomial $Q(t) = \|tx+(1-t)y\|_0^2(\lambda\cdot a(t)-b)$ is nonnegative for all t and vanishes for $t=0,1$. Hence $\lambda\cdot a(t) = b$ for all t and therefore $a(t)$, $0\leqq t\leqq 1$, is the line segment joining $a(x)$ to $a(y)$. Since $a(t)\in B$ for all t, the proof is complete.

Now, if $(x,y)_0 = -1$ we use the fact that $a(x) = a(-x)$ and replace x by $-x$; then $(-x,y)_0 = 1 \neq -1$ and hence the previous argument applies.

Joining together Proposition 2.1 and Lemma 2.3 gives us the main result of this section.

THEOREM 2.1. *Let X be a linear space and suppose $\|x\|_i^2$, $i=0,1,2$ are three Hilbert space semi-norms on X. Then*

$$\sup_{\substack{\|x\|_1\leqq 1\\ \|x\|_2\leqq 1}}\|x\|_0 = \min_{0\leqq\tau\leqq 1}\ \sup_{\tau\|x\|_1^2+(1-\tau)\|x\|_2^2\leqq 1}\|x\|_0.$$

Proof. According to Lemma 2.3 and Proposition 2.1

$$\inf_{a\in B}\|a\|_\infty = \max_{\lambda\in\Delta}\inf_{a\in B}\lambda\cdot a \tag{2.4}$$

and the maximum is taken on for some $\bar{\lambda}\in\Delta$. Since B lies in the first quadrant we have $\lambda\cdot a \leqq |\lambda|\cdot a$ for all $a\in B$. Thus the maximum in (2.4) is achieved for $\bar{\lambda}_1, \bar{\lambda}_2 \geqq 0$. Thus, reciprocating both sides of (2.4) yields the theorem.

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Theorem 2.1 says that for the set B, given by (2.3),

$$\inf_{a\in C(B)} \|a\|_\infty = \inf_{a\in B} \|a\|_\infty.$$

This result is not valid for every three choices of norms as the following example demonstrates.

Example 2.1. Let $x=(x_1, x_2)\in R^2$ and define

$$\|x\|_0^4 = 2x_1^4 + x_2^4, \qquad \|x\|_1^2 = 2x_1^2, \qquad \|x\|_2^2 = x_1^2 + x_2^2.$$

Note that only $\|x\|_0$ is *not* a Hilbert space semi-norm.

The set B in this case becomes

$$B = \{(t, t/2+\sqrt{1-t^2/2}) \mid 0 \leqq t \leqq \sqrt{2}\}.$$

The curve $f(t) = t/2+\sqrt{1-t^2/2}$ is an arc of an ellipse cut off from below by the line $1-\frac{1}{2}(\sqrt{2}-1)t$. Thus

$$\min_{a\in C(B)} \|a\|_\infty = \frac{2}{\sqrt{2}+1} < 1 = \min_{a\in B} \|a\|_\infty.$$

Remark. This example may be extended so that $\|x\|_0$ is a l^q-norm and $\|x\|_1$, $\|x\|_2$ are weighted l^p norms. When $q \geqq p$ it may be shown that Theorem 2.1 also fails in general.

It is evident from the proof of Lemma 2.3 that the nonnegativeness of the quadratic forms $\|x\|_1^2$, $\|x\|_2^2$ was not needed. Hence we also have

LEMMA 2.4. *Let C_1 and C_2 be real $n\times n$ matrices. Then the set*

$$C = \{((C_1x, x), (C_2x, x)) \mid (x, x) = 1\}$$

where

$$(x, y) = \sum_{i=1}^{n} x_i y_i$$

has a convex profile.

Lemma 2.4 yields the following result.

PROPOSITION 2.2.

$$\min_{(x,x)=1} \max_{i=1,2} (C_i x, x) = \max_{0\leqq\tau\leqq 1} \lambda(\tau C_1 + (1-\tau)C_2) \tag{2.5}$$

where $\lambda(C)$ = least eigenvalue of C.

Proof. Choose δ sufficiently large so that $C_1^\delta = C_1+\delta I$, $C_2^\delta = C_2+\delta I$ are both positive definite and let

$$C^\delta = \{((C_1^\delta x, x), (C_2^\delta x, x)) \mid (x, x) = 1\}.$$

Since C^δ has a convex profile and $0 \notin$ interior C^δ we have from Theorem 2.1

$$\min_{(x,x)=1} \max_{i=1,2} (C_i^\delta x, x) = \max_{0\leqq\tau\leqq 1} \min_{(x,x)=1} ((\tau C_1^\delta + (1-\tau)C_2^\delta)x, x)$$

which simplifies to (2.5).

Unfortunately this result and Theorem 2.1 do not extend to higher dimensions i.e. more than two matrices or Hilbert space semi-norms.

The following example demonstrates this fact.

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Example 2.2. Let $0<\alpha_0<\alpha_1$ and $X=R^2$. For $x=(x_1, x_2)$ we define

$$\|x\|_0^2=x_1^2+x_2^2, \qquad \|x\|_1^2=\alpha_0 x_1^2+\alpha_1 x_2^2,$$

$$\|x\|_2^2=\alpha_0\left(\frac{x_1-\sqrt{3}x_2}{2}\right)^2+\alpha_1\left(\frac{\sqrt{3}x_1+x_2}{2}\right)^2,$$

$$\|x\|_3^2=\alpha_0\left(\frac{x_1+\sqrt{3}x_2}{2}\right)^2+\alpha_1\left(\frac{\sqrt{3}x_1-x_2}{2}\right)^2.$$

Hence $\|x\|_2^2$, $\|x\|_3^2$ are successively determined by rotating the ellipse $\|x\|_1^2=1$ through 60° and 120°. If we parametrize the circle $\|x\|_0^2=1$ by $x=(\sin\theta, \cos\theta)$, $0\leqq\theta\leqq\pi$, then the set

$$B=\{(\|x\|_1^2, \|x\|_2^2, \|x\|_3^2) \mid \|x\|_0=1\}$$

becomes

$$B=\{(T(\theta), T(\theta+\pi/3), T(\theta+2\pi/3)) \mid 0\leqq\theta\leqq 2\pi\}$$

where $T(\theta)=\alpha_0\sin^2\theta+\alpha_1\cos^2\theta=(\alpha_0+\alpha_1)/2+((\alpha_1-\alpha_0)/2)\cos 2\theta$. Note that

$$\frac{1}{3}\left[T(\theta)+T\left(\theta+\frac{\pi}{3}\right)+T\left(\theta+\frac{2\pi}{3}\right)\right]=\frac{\alpha_0+\alpha_1}{2}$$

and consequently B and $C(B)$ lie on the plane

$$(\alpha_0+\alpha_1)/2=a_1+a_2+a_3, \qquad a=(a_1, a_2, a_3).$$

Thus

$$(\alpha_0+\alpha_1)/2\leqq\inf_{a\in C(B)}\|a\|_\infty.$$

Since $T(\theta+\pi)=T(\theta)$ we have

$$\frac{\alpha_0+\alpha_1}{2}(1, 1, 1)=\frac{1}{3}\sum_{i=1}^{3}\left(T(\theta_i), T\left(\theta_i+\frac{\pi}{3}\right), T\left(\theta_i+\frac{2\pi}{3}\right)\right),$$

where $\theta_1=0$, $\theta_2=\pi/3$, $\theta_3=2\pi/3$, and it follows that

$$\frac{\alpha_0+\alpha_1}{2}=\inf_{a\in C(B)}\|a\|_\infty.$$

This fact is not surprising since it may be easily verified that B is a circle with center $((\alpha_0+\alpha_1)/2)(1, 1, 1)$ and radius $((\alpha_1-\alpha_0)/2)\sqrt{3/2}$ in the plane described above.

However, it may be verified

$$\min_{a\in B}\|a\|_\infty=R^{-1}=\tfrac{1}{4}\alpha_0+\tfrac{3}{4}\alpha_1,$$

where R is the radius of the inscribed circle $x_1^2+x_2^2=R^2$ for the region $\|x\|_i^2\leqq 1$, $i=1, 2, 3$. Hence Theorem 2.1 fails in this case. Observe that $\partial C(B)=C(B)$ but B is not convex.

We end this section with some observations concerning extensions of Theorem 2.1 and Proposition 2.1 to higher dimensions.

Using Lemma 2.1 and Proposition 2.1 as our guide, we will say $A\subseteq R^n$ has a convex profile provided that $\partial C(A)\subseteq A$. Proposition 2.1 obviously remains valid in higher dimensions for sets with a convex profile.

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Let

$$B=\{(\|x\|_1^2,\cdots,\|x\|_n^2)|\ \|x\|_0^2=1, x\in X\}$$

where $\|x\|_i^2$, $i=1,\cdots,n$ are Hilbert space semi-norms over a linear space. In general we have seen that for $n\geqq 3$ B does not have a convex profile. However, the following extension of Lemma 2.3 is true:

$$\text{boundary } C(B)\subseteq D \tag{2.6}$$

where

$$D=\left\{a\,|\,a=\sum_{i=1}^{n-1}\mu_i a_i,\ a_i\in B,\ \mu_i\geqq 0,\ \sum_{i=1}^{n-1}\mu_i=1\right\},$$

where $n=2$, $D=B$.

To prove (2.6) we let $a_x=(\|x\|_1^2,\cdots,\|x\|_n^2)$, $a_0\in$ boundary $C(B)$ with $a_0=\sum_{i=1}^n\delta_i a_{x_i}$, $\delta_i>0$, $\sum_{i=1}^n\delta_i=1$ and $b=\lambda\cdot a_0\leqq\lambda\cdot a$ for all $a\in B$. Hence the hyperplane $\lambda\cdot a_0=\lambda\cdot a$ touches B at $a_{x_1},\cdots,a_{x_n}$. Thus the quadratic form $Q(t_1,\cdots,t_n)=\|\sum_{j=1}^n t_j\cdot x_j\|_0^2[\lambda\cdot a(t_1,\cdots,t_n)-b]$,

$$a(t_1,\cdots,t_n)=\left(\frac{\|\sum_{j=1}^n t_j\cdot x_j\|_1^2}{\|\sum t_j\cdot x_j\|_0^2},\cdots,\frac{\|\sum t_j\cdot x_j\|_n^2}{\|\sum t_j\cdot x_j\|_0^2}\right),$$

is nonnegative for all $t_1,\cdots,t_n$, and vanishes for $t_i=\delta_{ij}$ any j, $1\leqq j\leqq n$. Therefore $Q(t_1,\cdots,t_n)\equiv 0$ and $a(t_1,\cdots,t_n)\in B\cap H$, for $\|\sum t_jx_j\|_0\neq 0$ where H is the hyperplane $\lambda\cdot a_0=\lambda\cdot a$. Note as before if $\|t_1x_1+\cdots+t_nx_n\|_0=0$ then $\|-t_1x_1+\cdots+t_nx_n\|\neq 0$. Hence the curve $a(t_1,\cdots,t_n)$ connects the vectors $a_{x_1},\cdots,a_{x_n}$. The convex hull of this curve contains a_0 and is itself contained in an $n-1$ dimensional set. Hence $a_0\in D$, by Carathéodory's theorem.

Before we turn to our main application of Theorem 2.1 we will give another instance in which the conclusion of Lemma 2.1 remains valid for nonconvex sets.

PROPOSITION 2.3. *Let X be a linear space. Suppose $\|x\|_1,\cdots,\|x\|_n$ are semi-norms and Fx a linear functional defined on X. Then, for $A=\{(\|x\|_1,\cdots,\|x\|_n)|\ Fx=1\}$ we have*

$$\inf_{a\in A}\|a\|_\infty=\max_{a\in A}\min_{\lambda\in\Delta}\lambda\cdot a,\qquad \Delta=\left\{\lambda\,|\,\|\lambda\|_1=\sum_{j=1}^n|\lambda j|=1\right\}.$$

Proof. The essential observation which implies that $\inf_{a\in C(A)}\|a\|_\infty=\inf_{a\in A}\|a\|_\infty$ is that for any $a\in C(A)$ there is a $b\in A$ such that $a_j\geqq b_j$, $j=1,2,\cdots,n$. For if $a=\sum_{j=1}^m\mu_ja_{x_j}$, $a_{x_j}=(\|x_j\|_1,\cdots,\|x_j\|_n)$, $Fx_j=1$, $j=1,\cdots,m$, $\mu_j>0$, $\sum_{j=1}^m\mu_j=1$, then we may choose $b=a_x$ where $x=\sum_{j=1}^m\mu_jx_j$.

3. Optimal estimation of linear operators from inaccurate data. Referring to the optimal estimation scheme outlined in the Introduction, we assume here that the linear spaces X, Y, Z are equipped with inner product semi-norms, $K\subseteq X$ is of the form $K=\{x\in X|\ \|x\|_X\leqq 1\}$, $U:X\to Z$ is the linear operator to be estimated and $I:X\to Y$ is the information operator known within error $\varepsilon>0$.

Thus the problem is to determine a linear mapping A_0 (an algorithm) which minimizes

$$\sup_{\substack{\|Ix-y\|_Y\leqq 1\\ x\in K}}\|Ux-Ay\|_Z$$

over all mappings from $IK+\varepsilon S$, $S=\{y|\ \|y\|_Y\leqq 1\}$ into Z.

According to Theorem 2.1 there is a μ, $0 \leqq \mu \leqq 1$ such that

$$\sup_{\substack{\|x\|_X \leqq 1 \\ \|Ix\|_Y \leqq \varepsilon}} \|Ux\|_Z = \sup_{\mu\|x\|_X^2 + (1-\mu)\varepsilon^{-2}\|Ix\|_Y^2 \leqq 1} \|Ux\|_Z. \tag{3.1}$$

THEOREM 3.1. *For each fixed datum let $\bar{x} = \bar{x}(y)$ achieve*

$$m(y) = \min_{x \in X} ((1-\mu)\varepsilon^{-2}\|Ix - y\|_Y^2 + \mu\|x\|_X^2).$$

Then an optimal estimation algorithm exists and is given by the linear estimator A_μ defined by

$$A_\mu y = U\bar{x}$$

and moreover, for $\|x\|_X \leqq 1$, $\|Ix - y\| \leqq \varepsilon$

$$\|U\bar{x} - A_\mu y\|_Z \leqq \sqrt{1 - m(y)}\, e(K, \varepsilon).$$

In particular,

$$E(K, \varepsilon) = e(K, \varepsilon) = \sup_{\substack{\|x\|_X \leqq 1 \\ \|Ix\|_Y \leqq \varepsilon}} \|Ux\|_Z.$$

Note that $m(y) \leqq 1$ for any y for which there is an x, $\|x\|_X \leqq 1$, $\|Ix - y\|_Y \leqq \varepsilon$.

Proof. For any algorithm A and $0 \leqq \lambda \leqq 1$ we have from (1.1), (1.2)

$$\begin{aligned} e(K, \varepsilon) &\leqq \sup_{\substack{\|x\|_X \leqq 1 \\ \|Ix - y\|_Y \leqq \varepsilon}} \|Ux - Ay\|_Z \leqq \sup_{\lambda\|x\|_X^2 + (1-\lambda)\varepsilon^{-2}\|Ix - y\|_Y^2 \leqq 1} \|Ux - Ay\|_Z \\ &= R_\lambda(A). \end{aligned} \tag{3.2}$$

We wish to interpret $R_\lambda(A)$ as the error in an estimation problem. To this end, we introduce the product space $W = X \times Y$ with semi-norm

$$\|w\|_\lambda^2 = \lambda\|x\|_X^2 + (1-\lambda)\varepsilon^{-2}\|t\|_Y^2, \qquad w = (x, t), \tag{3.3}$$

a subset $\hat{K} = \{w \in W \mid \|w\|_\lambda \leqq 1\}$, an information operator $\hat{I}w = Ix - t$, and an operator $\hat{U}w = Ux$ to be estimated. $R_\lambda(A)$ is the error of the algorithm A in estimating $\hat{U}w$, $w \in \hat{K}$, from the *exact* information $\hat{I}w = y$ (a reflection of the fact that y is the exact value for a certain x and a certain error t). As mentioned in the Introduction, this is a situation in which it is known that there exists a linear algorithm which is optimal. We state this result as a lemma, see e.g. Micchelli and Rivlin [7].

LEMMA 3.1. *An optimal algorithm, A_λ, for estimating $\hat{U}w$ based on $\|w\|_\lambda \leqq 1$, $\hat{I}w = y$ is $A_\lambda y = \hat{U}\bar{w}$ where $\bar{w}$ achieves*

$$\min_{Iw = y} \|w\|_\lambda. \tag{3.4}$$

Moreover, for fixed y

$$\sup_{\|w\|_\lambda \leqq 1} \|\hat{U}w - A_\lambda y\|_Z \leqq \sqrt{1 - \|\bar{w}\|_\lambda^2} \sup_{\substack{\|w\|_\lambda \leqq 1 \\ \hat{I}w = 0}} \|\hat{U}w\|_Z. \tag{3.5}$$

We will give a proof of this lemma. But first we reinterpret (3.5) as applied to the product space W. We find that the optimal estimate A_λ for $\hat{U}w = Ux$ is $U\bar{x}$ where $\bar{x}$ achieves

$$\min_{x \in X} (\lambda\|x\|_X^2 + (1-\lambda)\varepsilon^{-2}\|Ix - y\|_Y^2). \tag{3.6}$$

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Moreover, the error of this algorithm and the intrinsic error for the estimation problem in W is given by (3.5)

$$R_\lambda(A_\lambda) = \sup_{\lambda\|x\|_X^2+(1-\lambda)\varepsilon^{-2}\|Ix\|_Y^2 \leqq 1} \|Ux\|_Z.$$

Now, for $\lambda = \mu$, it follows from (3.1) that

$$R_\mu(A_\mu) = \sup_{\substack{\|x\|_X \leqq 1 \\ \|Ix\|_Y \leqq \varepsilon}} \|Ux\|_Z = e(K, \varepsilon).$$

Hence with $\lambda = \mu$ all inequalities in (3.2) become equalities and in particular the algorithm A_μ is also an optimal algorithm for the original estimation problem.

Remark. The proof of Theorem 3.1 shows that in using the algorithm A_μ for the estimation problem of the theorem an error of exactly $e(K, \varepsilon)$ is committed. However, on using A_λ with $\lambda \neq \mu$ one still gets the fairly tight error bound

$$e(K, \varepsilon) \leqq \sup_{\substack{\|x\|_X \leqq 1 \\ \|Ix-y\| \leqq \varepsilon}} \|Ux - A_\lambda y\|_Z \leqq \max(\lambda^{-1/2}, (1-\lambda)^{-1/2}) e(K, \varepsilon).$$

For a similar observation see Morozov and Grebennikov [8] and Franklin [1].

Now, we return to Lemma 3.1 and for the convenience of the reader provide the proof of this result as given in [7].

Proof. Let $\|w\|_\lambda \leqq \|w\|_\lambda$ for all $\hat{I}w = y$. Then, since $\|\cdot\|$ is a Hilbert space seminorm, $\bar{w}$ depends linearly on y. Letting $A_\lambda \hat{y} = \hat{U}\bar{w}$ we prove (3.5) by noting that

$$\|w - \bar{w}\|_\lambda^2 = (w - \bar{w}, w - \bar{w})_\lambda = (w, w - \bar{w})_\lambda = (w, w)_\lambda - (\bar{w}, w)_\lambda,$$

since $\bar{w}$ is orthogonal to all w with $\hat{I}w = 0$, and $\hat{I}(w - \bar{w}) = 0$. Hence

$$\|w - \bar{w}_\lambda\|^2 \leqq 1 - \|\bar{w}\|_\lambda^2. \qquad \text{for } \|w\|_\lambda \leqq 1. \tag{3.7}$$

But

$$\|\hat{U}w - A_\lambda y\|_Z = \|\hat{U}(w - \bar{w})\|_\lambda$$

and therefore (3.5) follows directly from (3.7).

The minimum problem (3.6) has the interpretation of finding the element of the graph of I,

$$G = \{(x, Ix) \mid x \in X\},$$

which is closest to $(0, y)$ in the $\|\cdot\|_\mu$ norm. Suppose μ, defined by (3.1), is strictly between zero and one. Then the set G will be a closed subspace of $(W, \|\cdot\|_\mu)$ provided that I is a closed linear operator from X to Y. Thus in this case (3.6) achieves its minimum in X. Moreover, when $\|\cdot\|_X$ is a norm and I is a bounded linear operator with adjoint I^*: $Y \to X$, then the unique solution of (3.6) is easily seen to be

$$\bar{x}(y) = (I^*I + \rho\,\mathrm{id})^{-1} I^* y, \qquad \rho(1-\mu)^{-1}\varepsilon^2.$$

If $\mu = 0$ or 1 then the form of $x(y)$ changes. In the special case $Z = R$ it is possible to give a complete analysis of the cases $\mu \in (0, 1)$, $\mu = 0$ or 1 in terms of ε; see [7] for the details.

We present below some examples of Theorem 3.1. The first example is concerned with estimating the derivative of a smooth function from inaccurate function values.

Example 3.1. Let

$$W_2^n(-\infty,\infty)=\left\{f \mid \|f^{(n)}\|_2^2=\int_{-\infty}^{\infty}(f^{(n)}(x))^2\,dx<\infty\right\},$$

$$X=W_2^n(-\infty,\infty)\cap L^2(-\infty,\infty),\qquad K=\{f \mid f\in X, \|f^{(n)}\|_2\leqq 1\},$$

$$If=f,\qquad Uf=f^{(k)},\qquad Y=Z=L^2(R).$$

Thus the objective is to estimate $f^{(k)}$, optimally with respect to $\|\cdot\|_2$, from a knowledge of f known within ε and the fact that $\|f^{(n)}\|_2\leqq 1$.

First we evaluate the intrinsic error of the estimation problem and identify the optimal μ to use, i.e. find

$$\min_{0\leqq\lambda\leqq 1}\ \sup_{\lambda\|f^{(n)}\|_2^2+(1-\lambda)\varepsilon^{-2}\|f\|^2\leqq 1}\|f^{(k)}\|_2^2=\min_{\lambda}M(\lambda)=M(\mu).$$

Now

$$\begin{aligned}(1-\lambda)\varepsilon^{-2}\|f\|_2^2+\lambda\|f^{(n)}\|_2^2 &\\ =\frac{1}{2\pi}\int_{-\infty}^{\infty}&((1-\lambda)\varepsilon^{-2}|\omega|^{-2k}+\lambda|\omega|^{2(n-k)})|f^{(k)}(\omega)|^2\,d\omega\\ \geqq\frac{1-\lambda}{\varepsilon^2}\,\frac{n}{n-k}&\left(\frac{n-k}{k}\,\frac{\lambda\varepsilon^2}{1-\lambda}\right)^{k/n}\|f^{(k)}\|_2^2,\end{aligned}$$

since the factor multiplying the norm of $f^{(k)}$ furnishes a lower bound for $(1-\lambda)\varepsilon^{-2}t^{-k}+\lambda t^{n-k}$, $t>0$. Hence

$$M(k/n)=\min_{\lambda}M(\lambda)=\varepsilon^{2(1-k/n)}.$$

According to our theory then $\|f^{(k)}\|_2\leqq\varepsilon^{1-k/n}$, if $\|f^{(n)}\|_2\leqq 1$ and $\|f\|_2\leqq\varepsilon$. This, of course, says that

$$\|f^{(k)}\|_2\leqq\|f\|_2^{1-k/n}\|f^{(n)}\|_2^{k/n},$$

a version of Kolmogorov's inequality in $L^2(-\infty,\infty)$; see Tikhomirov [13, p. 117].

To obtain the optimal estimator of $f^{(k)}$ given g, $\|f-g\|\leqq\varepsilon$, we have to consider the minimum problem

$$\begin{aligned}\min_f\ \mu\|f^{(n)}\|_2^2+(1-\mu)\varepsilon^{-2}\|f-g\|_2^2&\\ =\min_f\frac{1}{2\pi}\int_{-\infty}^{\infty}&(\mu|\omega|^{-2n}|f(\omega)|^2+(1-\mu)\varepsilon^{-2}|\hat{f}(\omega)-\hat{g}(\omega)|)\,d\omega.\end{aligned}$$

Thus the solution is

$$(P_\rho g)(t)=\int_{-\infty}^{\infty}K_\rho(t-t')g(t')\,dt$$

where

$$\rho=\frac{\mu\varepsilon^2}{1-\mu}=\frac{k\varepsilon^2}{n-k},\qquad K_\rho(t)=\frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{e^{i\omega t}}{1+\rho\omega^{2n}}\,d\omega,$$

and the optimal estimator for $f^{(k)}$ is $(P_\rho g)^{(k)}$.

It is interesting to note that if instead we wish to estimate $f^{(k)}$ only at a single point, say $f^{(k)}(0)$, with the same given information, then $(P_\rho g)^{(k)}(0)$ again furnishes an optimal estimator but with a *different* value of ρ. To obtain this value of ρ we begin by

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noting that $K_\rho(t-t')$ is the reproducing kernel for the Hilbert space X with norm $|f|_\rho^2 = \|f\|_2^2 + \rho\|f^{(n)}\|_2^2$. Hence if

$$(f, g)_\rho = (f, g)_2 + \rho(f^{(n)}, g^{(n)})_2 \quad \text{then} \quad f(x) = (K_\rho(t-\cdot), f(\cdot))_\rho, \qquad f \in X.$$

Therefore

$$\begin{aligned}
&\sup_{\lambda\|f^{(n)}\|_2^2 + (1-\lambda)\varepsilon^{-2}\|f\|_2^2 \leqq 1} |f^{(k)}(0)|^2 \\
&\qquad = (1-\lambda)^{-1}\varepsilon^2 \sup_{|f|_\rho \leqq 1} |f^{(k)}(0)|^2 \\
&\qquad = (1-\lambda)^{-1}\varepsilon^2 |K_\rho^{(k)}(0-\cdot)|_\rho^2 \\
&\qquad = (1-\lambda)^{-1}\varepsilon^2(-1)^k K_\rho^{(2k)}(0) \\
&\qquad = (1-\lambda)^{-1}\varepsilon^2\rho^{-(2k+1)/(2n)}\frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{\omega^{2k}}{1+\omega^{2n}}\,d\omega \\
&\qquad = \lambda^{(2k+1)/(2n)} \\
&\qquad\quad \cdot (1-\lambda)^{-1+(2k+1)/(2n)}\varepsilon^{(2(n-k)-1)/n}\frac{1}{2n\sin\pi(2k+1)/(2n)}.
\end{aligned}$$

Thus the intrinsic error of estimation is obtained by minimizing this function of λ for $0<\lambda<1$. The minimum is achieved for $\lambda = \mu = k/n + 1/(2n)$ and the value of the minimum is

$$\left(\frac{2n}{2k+1}\right)^{(2k+1)/(2n)}\left(\frac{2(n-k)-1}{2n}\right)^{-(2(n-k)-1)/(2n)}\varepsilon^{(2(n-k)-1)/n}\left(2n\sin\pi\frac{2k+1}{2n}\right)^{-1}.$$

Hence the optimal value for ρ is $\rho = (k+\frac{1}{2})/(n-k+\frac{1}{2})\varepsilon^2$ which may be compared to $\rho = k/(n-k)\varepsilon^2$, the optimal value when estimating the *full* kth derivative. Thus we have also shown that

$$\begin{aligned}
|f^{(k)}(0)| \leqq &\left(\frac{2n}{2k+1}\right)^{(2k+1)/(4n)}\left(\frac{2n}{2(n-k)-1}\right)^{(2(n-k)-1)/(2n)}\left(2n\sin\frac{\pi(2k+1)}{2n}\right)^{-1/2} \\
&\cdot \|f\|_2^{(2(n-k)-1)/(2n)}\|f^{(n)}\|_2^{(2k+1)/(2n)}
\end{aligned}$$

for $k<n$. Since the norms above are translation invariant we may replace $f(x)$ by $f(x+x_0)$ above and thus obtain a sharp bound of $\|f^{(k)}\|_\infty$ in terms of $\|f\|_2, \|f^{(n)}\|_2$; see also Tikhomirov [13; p. 120].

Example 3.2. *Smoothing of data.* Here we choose $X = W_2^n[-1,1]$, $K = \{f \mid f \in X, \|f^{(n)}\|_2^2 = \int_{-1}^{1}(f^{(n)}(x))^2\,dx \leqq 1\}$, $Y = E^N$, $Z = L^2[-1,1]$ and $Uf = f$, $If = (f(x_1), \cdots, f(x_N))$, $-1 \leqq x_1 < \cdots < x_N \leqq 1$, $N \geqq n$. Thus we wish to estimate f, given $\|f^{(n)}\|_2 \leqq 1$ and $f(x_1), \cdots, f(x_N)$ known with error ε. The optimal estimator is determined by

$$\min_f \rho\|f^{(n)}\|_2^2 + \sum_{i=1}^{N}(f(x_i)-y_i), \qquad \rho = \frac{\lambda}{1-\lambda}\varepsilon^2.$$

It is known that the unique solution to this problem is a natural spline of degree $2n-1$ with knots at $x_1 < \cdots < x_N$. It is difficult to determine the exact value of the optimal parameter in this case. Although, Theorem 3.1 demonstrates that it is the value at which the convex function

$$\sup_{(1-\lambda)\varepsilon^{-2}\sum_{i=1}^{n}(f(x_i)-y_i)^2 + \lambda\|f^{(n)}\|_2^2} \|f\|_2$$

attains its minimum in (0, 1). For other methods for selecting λ see Reinsch [9], [10], Wahba and Wold [14], [15] and Wahba [16].

Example 3.3. *Numerical differentiation of inaccurate data.* Let $X = \bar{W}_2^n[-\pi, \pi] = \{f \mid f, 2\pi\text{-periodic}, f^{(n)} \in L^2[-\pi, \pi]\}$ with inner product

$$(f, g) = \frac{1}{2\pi} \int_{-\pi}^{\pi} (f(t)g(t))\, dt,$$

$K = \{f \mid f \in X, \|f^{(n)}\|_2 \leqq 1\}$ and the operator to be estimated $Uf = f^{(k)}$, $Z = L^2[-\pi, \pi]$. For the information operator we take (1) $If = f$, $Y = L^2[-\pi, \pi]$ and also (2) $If = (f(x_1), \cdots, f(x_N))$, $Y = E^N$.

(1) Since f is periodic we may use the Fourier series expansion as we used the Fourier transform in Example 3.1 and show that $\mu = k/n$ is optimal in this case as well. The optimal algorithm estimates $f^{(k)}$ from the given g, $\|g - f\| \leqq \varepsilon$, by $\bar{f}^{(k)}$ where $\bar{f}$ achieves

$$\min_{f \in X} \|f - g\|_2^2 + \frac{k\varepsilon^2}{n-k} \|f^{(n)}\|_2^2.$$

It is easily seen that

$$\bar{f}(t) = \int_{-\pi}^{\pi} R_\rho(t - t')g(t')\, dt', \qquad R_\rho(t) = \sum_{-\infty}^{\infty} K_\rho(t + 2\pi j), \qquad \rho = \frac{k\varepsilon^2}{n-k},$$

where K_ρ is defined in Example 3.1.

(2) When the information is discrete,

$$If = (f(x_1), \cdots, f(x_N)), \qquad x_j = -\pi + \frac{j}{N} 2\pi, \qquad j = 1, \cdots, N$$

with

$$\frac{1}{N} \sum_{i=1}^{N} (f(x_i) - y_i)^2 \leqq \varepsilon^2,$$

it is no longer possible to calculate the optimal μ_N explicitly. Let us therefore estimate the value of μ_N asymptotically as $N \to \infty$ from its continuous analogue in (1). For any fixed λ we compare

$$E(\lambda) = \min_{\|f^{(k)}\| = 1} \lambda \|f^{(n)}\|^2 + (1 - \lambda)\varepsilon^{-2} \|f\|^2$$

and

$$E_N(\lambda) = \min_{\|f^{(k)}\| = 1} \lambda \|f^{(n)}\|^2 + (1 - \lambda)\varepsilon^{-2} \frac{1}{N} \sum_{i=1}^{N} |f(x_i)|^2.$$

Now, for some constant C

$$\left| \|f\|^2 - \frac{1}{N} \sum_{i=1}^{N} |f(x_i)|^2 \right| \leqq CN^{-2k} \|f^{(k)}\|^2,$$

from which it follows that

$$|E_N(\lambda) - E(\lambda)| \leqq C\varepsilon^{-2} N^{-2k} = \delta_N. \tag{3.8}$$

This bounds the concave function $E_N(\lambda)$ between the concave functions $E(\lambda) \pm \delta_N$.

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We have already verified that $E(\lambda)$ achieves its maximum at k/n and $E(k/n)=\varepsilon^{-2(1-k/n)}$. If $E_N(\lambda)$ achieves its maximum at μ_N then this value must lie in the interval given by the intersections $\lambda_\pm$ of the horizontal line $y=E(k/n)-\delta_N$ with the upper envelope $E(\lambda)+\delta_N$. We expand by Taylor's formula,

$$E(\lambda)=E(k/n)+\tfrac{1}{2}(k/n-\lambda)^2E''(\xi)$$

and observe that $|E''(\xi)|>d>0$, whence

$$2\delta_N=E(k/n)-E(\lambda_\pm)\geqq\tfrac{1}{2}(k/n-\lambda_\pm)^2\,d.$$

This together with (3.7) shows that

$$\left|\mu_N-\frac{k}{n}\right|\leqq\frac{C(\varepsilon,k,n)}{N^k}.$$

Rather than give more examples of Theorem 3.1 we turn instead to another application of Theorem 2.1.

4. *n*-widths. Let $\mathscr{A}$ be a subset of a normed linear space. The n-width of $\mathscr{A}$ is defined as

$$d_n(\mathscr{A};X)=d_n(\mathscr{A})=\inf_{X_n}\sup_{x\in\mathscr{A}}\inf_{y\in X_n}\|x-y\|$$

where the infimum is taken over all n-dimensional subspaces of X.

It is possible to compute the n-width of sets $\mathscr{A}$ determined by a single quadratic form, i.e. ellipsoidal sets. For instance, it may be shown that if K is a compact operator taking a Hilbert space X into a Hilbert space Y and

$$K^TK\varphi_i=\lambda_i\varphi_i,\qquad(\varphi_i,\varphi_j)=\delta_{ij},\qquad i,j=0,1,\cdots,$$
$$\lambda_0\geqq\lambda_1\geqq\cdots\geqq\lambda_n\to0,$$

then $d_n(\mathscr{K})=\lambda_n^{1/2}$ where $\mathscr{K}=\{Kx|\ \|x\|\leqq1\}$. Moreover, for $X_n^0=[\varphi_0,\cdots,\varphi_{n-1}]$, the linear subspace spanned by $\varphi_0,\cdots,\varphi_{n-1}$ we have

$$\sup_{x\in\mathscr{A}}\inf_{y\in X_n^0}\|x-y\|=d_n(\mathscr{A})$$

i.e., X_n^0 is an optimal subspace for $\mathscr{A}$.

Now, we may use Theorem 2.1 to give a general method of computing n-widths of sets determined by *two* quadratic forms.

Let X be a linear subspace with three Hilbert space semi-norms $\|x\|_i$, $i=0,1,2$. We define

$$\mathscr{A}=\{Kx|\ \|x\|_i\leqq1,\ i=1,2\}$$

where K is a linear operator mapping X into itself and consider the problem of computing the n-width of $\mathscr{A}$ as a subset of $(X,\|\cdot\|_0)$.

THEOREM 4.1.

$$d_n(\mathscr{A})=\inf_{0\leqq\lambda\leqq1}d_n(\mathscr{A}_\lambda)$$

where $\mathscr{A}_\lambda=\{Kx|\ \lambda\|x\|_1^2+(1-\lambda)\|x\|_2^2\leqq1\}$. *Moreover, if* $d_n(\mathscr{A}_\mu)=d_n(\mathscr{A})$ *and* X_n^0 *is an optimal subspace for* $\mathscr{A}_\mu$ *then it is also an optimal subspace for* $\mathscr{A}$.

Proof. Let X_n be an n-dimensional subspace of X. Suppose $P_n: X \to X_n$ is the orthogonal projection of X onto X_n. Then $\|Q_nKx\|_0 = \inf_{y \in X_n} \|Kx - y\|_0$ where $Q_nx = x - P_nx$. Since $\|Q_nKx\|_0$ is also a Hilbert space semi-norm on X we have from Theorem 2.1

$$\min_{0 \leqq \lambda \leqq 1} \sup_{\lambda \|x\|_1^2 + (1-\lambda)\|x\|_2^2 \leqq 1} \|Q_nKx\|_0 = \sup_{\|x\|_i \leqq 1,\, i=1,2} \|Q_nKx\|_0.$$

Hence

$$\begin{aligned} d_n(\mathscr{A}) &= \inf_{X_n} \sup_{Kx \in \mathscr{A}} \|Q_nKx\|_0 \\ &= \inf_{X_n} \min_{0 \leqq \lambda \leqq 1} \sup_{Kx \in \mathscr{A}} \|Q_nKx\|_0 \\ &= \inf_{0 \leqq \lambda \leqq 1} \inf_{X_n} \sup_{Kx \in \mathscr{A}_\lambda} \|Q_nKx\|_0 = \inf_{0 \leqq \lambda \leqq 1} d_n(\mathscr{A}_\lambda). \end{aligned}$$

Now, if $d_n(\mathscr{A}_\mu) = \sup_{Kx \in \mathscr{A}_\mu} \|Q_n^0Kx\| = d_n(\mathscr{A})$ for some n-dimensional subspace X_n^0 then

$$\sup_{Kx \in \mathscr{A}_\mu} \|Q_n^0Kx\|_0 \leqq \sup_{Kx \in \mathscr{A}_\lambda} \|Q_n^0Kx\|_0 \quad \text{for all } \lambda, \quad 0 \leqq \lambda \leqq 1.$$

Thus

$$\begin{aligned} d_n(\mathscr{A}) = \sup_{Kx \in \mathscr{A}_\mu} \|Q_n^0Kx\| &= \min_{0 \leqq \lambda \leqq 1} \sup_{Kx \in \mathscr{A}_\lambda} \|Q_n^0Kx\| \\ &= \sup_{Kx \in \mathscr{A}} \|Q_n^0Kx\|, \end{aligned}$$

that is, X_n^0 is an optimal subspace for $\mathscr{A}$.

A similar result holds for the Gel'fand n-width of $\mathscr{A}$,

$$d^n(\mathscr{A}) = \inf_{L_n} \sup_{x \in \mathscr{A} \cap L_n} \|x\|_0 = \inf_{0 \leqq \lambda \leqq 1} d^n(\mathscr{A}_\lambda),$$

where the minimum is taken over all subspaces L_n of X of codimension n.

We demonstrate the usefulness of Theorem 4.1 with two examples.

Example 4.1. *n-width of the space of essentially time-limited and band-limited signals.* Let $X = \mathscr{L}_2(-\infty, \infty)$, the space of complex valued square integrable functions on the real line with inner product

$$(f, g) = \int_{-\infty}^{\infty} \bar{f}(t) g(t)\, dt, \qquad \|f\|^2 = (f, f).$$

Denote by $\mathscr{D}$ the space of functions $f \in \mathscr{L}_2(-\infty, \infty)$ vanishing outside $(-\tau, \tau)$ and by $\mathscr{B}$ those functions whose Fourier transform vanishes outside $(-\sigma, \sigma)$. Let D, B be the projections of $\mathscr{L}_2(R)$ onto these spaces, i.e.

$$Df(t) = \begin{cases} f(t), & |t| \leqq \tau, \\ 0, & |t| > \tau, \end{cases}$$

$$Bf(t) = \int_{-\infty}^{\infty} f(t') \frac{\sin 2\pi\sigma(t-t')}{\pi(t-t')}\, dt'.$$

We want to compute the n-widths with respect to $\mathscr{L}_2(-\infty, \infty)$ of the class

$$\mathscr{A} = \{f \mid f \in \mathscr{L}_2(-\infty, \infty) \|(1-D)f\| \leqq \varepsilon, \|(1-B)f\| \leqq \eta\},$$

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the class of functions which are time-limited at level ε and band-limited at level η. The importance of this class of functions and the motivation for computing the n-width of $\mathscr{A}$ are discussed in Slepian [11]. Theorem 4.1 gives us a means for computing the n-width of $\mathscr{A}$. First we compute the n-widths of

$$\mathscr{A}_\lambda = \left\{ f \mid f \in \mathscr{L}_2(-\infty, \infty), \frac{\lambda}{\varepsilon^2}\|(1-D)f\|^2 + \frac{1-\lambda}{\eta^2}\|(1-B)f\|^2 \leqq 1 \right\}$$

and then minimize the result over λ (for another treatment see Slepian [11], Melkman [5]). As we already pointed out, the n-width of $\mathscr{A}_\lambda$ is simply the square root of the $(n+1)$st eigenvalue corresponding to the variational equation

$$\frac{1}{\mu} f = \frac{\lambda}{\varepsilon^2}(1-D)f + \frac{1-\lambda}{\eta^2}(1-B)f. \tag{4.1}$$

In order to analyze this equation we need the following facts, Slepian and Pollak [12]. Let $\{\lambda_i\}_0^\infty$, $\{\psi_i(t)\}_0^\infty$ be the complete set of eigenvalues and orthonormal eigenfunctions of the completely continuous symmetric kernel BD,

$$\begin{aligned} BD\psi_i(t) &= \int_{-T}^{T} \frac{\sin 2\pi\sigma(t-t')}{\pi(t-t')} \psi_i(t')\, dt' \\ &= \lambda_i \psi_i(t'), \qquad |t'| \leqq T. \end{aligned}$$

It is known that

$$1 > \lambda_0 > \lambda_1 > \cdots > \lambda_j \to 0, \qquad j \to \infty.$$

Set

$$\frac{1}{\sqrt{\lambda_i}} D\psi_i = \varphi_i.$$

Then

$$\begin{aligned} &\|\varphi_i\| = 1, \qquad \psi_i \in \mathscr{B}, \qquad \varphi_i \in \mathscr{D}, \\ &D\psi_i = \sqrt{\lambda_i}\varphi_i, \qquad B\varphi_i = \sqrt{\lambda_i}\psi_i \end{aligned}$$

and $\{\varphi_i\}_0^\infty$ is complete in $\mathscr{D}$ while $\{\psi_i\}_0^\infty$ is complete in $\mathscr{B}$.

Returning to the variational equation, since $Df \in \mathscr{D}$ it may be expanded in the $\{\varphi_i\}$ while Bf may be expanded in the ψ_i. Hence (4.1) becomes

$$\left(\frac{1}{\mu} + \frac{\lambda}{\varepsilon^2} + \frac{1-\lambda}{\eta^2}\right) f = \frac{\lambda}{\varepsilon^2} \sum_{i=0}^{\infty} (f, \varphi_i)\varphi_i + \frac{1-\lambda}{\eta^2} \sum_{i=0}^{\infty} (f, \psi_i)\psi_i.$$

Multiplying this equation with φ_j and then with ψ_j and integrating gives

$$\left(-\frac{1}{\mu} + \frac{\lambda}{\eta^2}\right)(f, \varphi_j) - \frac{(1-\lambda)}{\eta^2}\sqrt{\lambda_j}(f, \psi_j) = 0$$

and

$$-\frac{\lambda}{\varepsilon^2}\sqrt{\lambda_j}(f, \varphi_j) + \left(-\frac{1}{\mu} + \frac{1-\lambda}{\varepsilon^2}\right)(f, \psi_j) = 0, \qquad j = 0, 1, 2, \cdots.$$

These equations have a nontrivial solution if and only if

$$\frac{1}{\mu} = \frac{1}{2}\left[\frac{\lambda}{\varepsilon^2} + \frac{1-\lambda}{\eta^2} \pm \sqrt{\left(\frac{\lambda}{\varepsilon^2} + \frac{1-\lambda}{2}\right)^2 - \frac{4\lambda(1-\lambda)}{\varepsilon^2\eta^2}(1-\lambda_j)}\right]$$

for some μ. It is easily seen that the minus sign always yields a value of μ which is larger than any of the + sign values. Hence the nth eigenvalue is obtained by letting $j=n$.

The minimum of $\mu_n(\lambda)$ is obtained for

$$\lambda^* = \frac{\varepsilon^2+\varepsilon\eta\sqrt{\lambda_n}}{\varepsilon^2+\eta^2+2\varepsilon\eta\sqrt{\lambda_n}}$$

and

$$d_n(\mathscr{A};\mathscr{L}_2(R)) = d_n(\mathscr{A}_{\lambda^*};\mathscr{L}_2(R) = \frac{\varepsilon^2+\eta^2+2\varepsilon\eta\sqrt{\lambda_n}}{1-\lambda_n}.$$

In particular, the case $n=0$

$$\sup_{f\in\mathscr{A}} \|f\|^2 = \frac{\varepsilon^2+\eta^2+2\varepsilon\eta\sqrt{\lambda_0}}{1-\lambda_0}$$

yields the shape of the boundary of the planar set

$$A = \{(a_1, a_2) = (\|(1-D)f\|^2, \|(1-B)f\|^2) \mid \|f\| = 1\},$$

namely

$$\frac{(\sqrt{a_1}+\sqrt{a_2})^2}{1-\sqrt{\lambda_0}} + \frac{(\sqrt{a_1}-\sqrt{a_2})^2}{1+\sqrt{\lambda_0}} = 2$$

which is a convex curve as is to be expected by Lemma 2.3.

Example 4.2. *Approximation of smooth functions with bounded extensions.* Given T, $0<T<1$ we define

$$\mathscr{H}_2^1(-\infty,\infty) = L^2(-\infty,\infty)\cap W_2^1(-\infty,\infty)$$

and

$$\mathscr{S} = \left\{ f\in\mathscr{H}_2^1 \,\middle|\, \int_{T<|t|} f^2(t)\,dt \leqq \varepsilon^2, \int_{-\infty}^{\infty} (f'(t))^2\,dt \leqq 1 \right\}.$$

We will compute the n-width of the set $\mathscr{S}$ as a subset of $L^2(-T,T)$.

For $n=0$ this leads us to the "concentration problem" for the Sobolev class $\mathscr{S}$, that is

$$\sup_{f\in\mathscr{S}} \int_{-T}^{T} f^2(t)\,dt.$$

The solution of this problem may be interpreted as the smooth function whose size outside $(-T,T)$ is "small" while it is most concentrated in $(-T,.T)$. The concentration problem for various classes of entire functions is studied in Slepian, Pollak [12] and Logan [3].

Upper and lower bounds for $d_n(\mathscr{S})$ may be determined by noting that

$$B\cap X_{n+1} \subseteq \mathscr{S} \subseteq U_2^1(-T,T) = \left\{ f\in W_2^1(-T,T) \,\middle|\, \int_{-T}^{T} |f'(t)|^2\,dt \leqq 1 \right\}$$

where

$$B = \left\{ f \,\middle|\, \int_{-T}^{T} f^2(t)\,dt \leqq \frac{T^2}{(n+1)^2\pi^2} \right\}$$

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and

$$X_{n+1}=\left\{\sum_{j=1}^{n+1} a_j \sin\frac{j\pi x}{T} \;\middle|\; (a_1,\cdots,a_{n+1})\in R^{n+1}\right\}.$$

Thus for all $\varepsilon \geqq 0$

$$\frac{T}{(n+1)\pi}\leqq d_n(S)\leqq\frac{T}{n\pi},\qquad n=1,2,\cdots.$$

Here we have used the well-known facts that

$$d_n(B\cap X_{n+1};L^2(-T,T))=\frac{T}{(n+1)\pi},\qquad d_n(U_2^1(-T,T);L^2(-T,T))=\frac{T}{n\pi}.$$

Now, to determine the exact value of $d_n(\mathcal{S})$ we are led by Theorem 4.1 to the eigenvalue problem,

$$\frac{1}{\mu}\chi_T f=\lambda\varepsilon^{-2}(1-\chi_T)f-(1-\lambda)f'',\qquad f\in L^2(-\infty,\infty)\cap C^1(-\infty,\infty),$$

where χ_T = the characteristic function of $(-T,T)$. An easy calculation shows that the eigenfunctions are

$$f(x)=\begin{cases} e^{-\sqrt{\delta}(x-T)}, & x\geqq T,\\ \dfrac{\cos\sqrt{\rho}x}{\cos\sqrt{\rho}T}, & 0\leqq x\leqq T,\end{cases}\qquad f(x)=f(-x),$$

where

$$\delta=\frac{\lambda}{\varepsilon^2(1-\lambda)}\quad\text{and}\quad\rho=\frac{1}{\mu(1-\lambda)}.$$

The eigenvalues are determined by the equation

$$\sqrt{\frac{\mu\lambda}{\varepsilon^2}}=\tan\frac{T}{\sqrt{\mu(1-\lambda)}},$$

the condition which insures that f' is continuous at T. If we denote by $\mu_0(\lambda)<\cdots<\mu_n(\lambda)<\cdots$ the positive roots of this equation then

$$d_n(\mathcal{S})=\min_{0\leqq\lambda\leqq 1}\mu_n^{1/2}(\lambda).$$

Letting $x_0(t)<\cdots<x_n(t)<\cdots$ be the positive roots of

$$x\tan x=t,\qquad n\pi<x_n(t)<n\pi+\frac{\pi}{2},$$

we may express the eigenvalues in the form

$$\mu_n^{-1/2}(\lambda)=\frac{x_n(t)}{\sqrt{T^2+\varepsilon^2t^2}},\quad\text{where}\quad t=\frac{T}{\varepsilon}\sqrt{\frac{\lambda}{1-\lambda}}.$$

Thus

$$d_n(\mathcal{S})=\min_{t\geqq 0}\frac{\sqrt{T^2+\varepsilon^2t^2}}{x_n(t)}=x_n^{-1}\sqrt{T^2+\varepsilon^2\sigma_n^2}$$

where (σ_n, x_n) is given uniquely by the equations

$$x_n \tan x_n = \sigma_n, \qquad \pi n < x_n < \pi n + \frac{\pi}{2}, \qquad \sigma_n^3 + \sigma_n x_n^2 = \left(\frac{T}{\varepsilon}\right)^2.$$

5. A final example. In this section, we present another example to which our methods apply. It is difficult to present a general result of any interest which encompasses this example. However, it does serve to suggest other cases which may be solved by the approach we have used in §§ 3 and 4.

The example concerns estimating a smooth periodic function f from inaccurate Fourier coefficients. Thus we choose $X = \bar{W}_2^n[-\pi, \pi]$, $K = \{f \mid \|f^{(n)}\|^2 = \int_{-\pi}^{\pi} (f^{(n)}(x))^2\, dx \leqq 1\}$, $Z = L^2[-\pi, \pi]$, $Uf = f$, and $If = (\hat{f}_{-J}, \cdots, \hat{f}_J)$ where

$$\hat{f}_j = \frac{1}{\sqrt{2\pi}} \int_{-\pi}^{\pi} f(t)\, e^{ijt}\, dt.$$

This problem is solved in [7], where an L^2-norm is used to measure the error in the Fourier coefficients. Here we are interested in the case that $|f_j - y_j| \leqq \varepsilon_j$, $|j| \leqq J$, $\varepsilon_j > 0$. Now, as before, the error in this estimation problem cannot be less than

$$\sup_{\substack{|\hat{f}_j| \leqq \varepsilon_j, |j| \leqq J \\ \|f^{(n)}\| \leqq 1}} \|f\|.$$

An upper bound on the error is determined in the following manner. Given any $\mu > 0$, $\lambda_j \geqq 0$, $|j| \leqq J$, with $\mu + \sum_{|j| \leqq J} \lambda_j = 1$, we let $\bar{f} = \bar{f}(y)$ solve

$$\min_f \mu \|f^{(n)}\|^2 + \sum_{|j| \leqq J} \lambda_j \varepsilon_j^{-2} (\hat{f}_j - y_j)^2.$$

An easy calculation shows (where y_j are the given approximate Fourier coefficients) that

$$\hat{\bar{f}}_j = \begin{cases} \dfrac{\lambda_j y_j}{\mu \varepsilon_j^2 |j|^{2n} + \lambda_j}, & |j| \leqq J, \\ 0, & |j| \geqq J \end{cases}$$

(if $\lambda_0 = 0$ then $\bar{f}_0$ is arbitrary). Following the method used in the proof of Theorem 3.1 we see that

$$\begin{aligned} \sup_{\substack{|\hat{f}_j| \leqq \varepsilon_j \\ \|f^{(n)}\| \leqq 1}} \|f\| &\leqq \sup_{\substack{|\hat{f}_y - y_j| \leqq \varepsilon_j \\ \|f^{(n)}\| \leqq 1}} \|f - \bar{f}(y)\| \\ &\leqq \sup_{\mu \|f^{(n)}\|^2 + \sum_{|j| \leqq J} \lambda_j \varepsilon_j^{-2} |\hat{f}_j - y_j|^2 \leqq 1} \|f - \bar{f}(y)\| \\ &= \sup_{\mu \|f^{(n)}\|^2 + \sum_{|j| \leqq J} \lambda_j \varepsilon_j^{-2} |\hat{f}_j|^2 \leqq 1} \|f\|. \end{aligned}$$

To complete the example it remains to prove that there exists $\mu > 0$, $\lambda_j \geqq 0$, $\mu + \sum_{|j| \leqq J} \lambda_j = 1$ so that the right hand side of the above inequality equals the left. Thus we wish to show that

$$\inf_{a \in A} \|a\|_\infty = \min_{\lambda \in \Delta} \max_{a \in A} \lambda \cdot a \tag{5.1}$$

where

$$A = \{(\|f^{(n)}\|^2, \varepsilon_{-J}^{-2} |\hat{f}_{-J}|^2, \cdots, \varepsilon_J^{-2} |\hat{f}_J|^2) \mid \|f\| = 1\}$$

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and

$$\Delta = \left\{ \lambda = (\mu, \lambda_{-J}, \cdots, \lambda_J) \mid \mu > 0, \lambda_j \geqq 0, \mu + \sum_{|j| \leqq J} \lambda_j = 1 \right\}.$$

Using the Fourier series representation of f we see that A is identical with the set

$$\left\{ \left(\sum_{-\infty}^{\infty} j^{2n} a_j, \varepsilon_{-J}^{-2} a_{-J}, \cdots, \varepsilon_J^{-2} a_J \right) \middle| \sum_{-\infty}^{\infty} a_j = 1, a_j \geqq 0 \right\}.$$

This representation of A reveals that it is a *convex* set and hence (5.1) is valid.

REFERENCES

[1] J. N. FRANKLIN, *On Tikhonov's method for ill-posed problems*, Math. Comput., 28 (1974), pp. 889–907.

[2] M. GOLOMB AND H. F. WEINBERGER, *Optimal approximation*, On Numerical Approximation, R. E. Langer, ed., University of Wisconsin Press, Madison, 1959, pp. 117–190.

[3] B. LOGAN, JR., *Properties of high-pass signals*, Ph.D. dissertation, Columbia University, 1965.

[4] A. G. MARCHUK AND K. YU. OSIPENKO, *Best approximation of functions specified with an error at a finite number of points*, Math. Notes, 17 (1975), pp. 359–368.

[5] A. A. MELKMAN, *n-widths and optimal interpolation of time- and band-limited functions*, Optimal Estimation in Approximation Theory, C. A. Micchelli, T. J. Rivlin, eds., Plenum Press, 1977, pp. 55–68.

[6] C. A. MICCHELLI, *Optimal estimation of linear functionals*, IBM Research Report No. 5729, 1975.

[7] C. A. MICCHELLI AND T. J. RIVLIN, *A survey of optimal recovery*, Optimal Estimation in Approximation Theory, C. A. Micchelli, T. J. Rivlin, eds., Plenum Press, 1977, pp. 1–53.

[8] V. A. MOROZOV AND GREBENNIKOV, *On optimal approximations of operators*, Soviet Math. Dokl., 16 (1975), *pp.* 1084–1088.

[9] C. REINSCH, *Smoothing by spline functions*, Numer. Math., 10 (1967), pp. 177–183.

[10] ———, *Smoothing by spline functions II*, Ibid., 16 (1971), pp. 451–454.

[11] D. SLEPIAN, *On bandwidth*, Proc. IEEE, 64 (1976), pp. 292–300.

[12] D. SLEPIAN AND H. O. POLLACK, *Prolate spheroidal wave-functions, Fourier analysis and uncertainty, I*, Bell System Tech. J., 40 (1961), pp. 43–64.

[13] V. M. TIKHOMIROV, *Nekotorye Voprosy Teorii Priblijehiĭ*, Moscow University, 1976. (In Russian.)

[14] GRACE WAHBA AND SVANTE WOLD, *A completely automatic French curve: Fitting spline functions by cross-validation*, Comm. Statist., 4 (1975), pp. 1–17.

[15] ———, *Periodic splines for spectral density estimation: The use of cross-validation for determining the degree of smoothing*, Ibid., 4 (1975), pp. 125–141.

[16] GRACE WAHBA, *A Survey of some smoothing problems and the method of generalized cross-validation for solving them*, Rep. TR #430, Statistics Department, University of Wisconsin-Madison, 1976, to appear; Proceedings of the Conference on the Applications of Statistics (Dayton, Ohio, June 14–18, 1976), P. R. Krishnaiah, ed.

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